MIL-R-27777/4(EC)
31 October 1968

MILITARY SPECIFICATION SHEET

RELAY, TELEGRAPH, PASSIVE, SOLID STATE

RELAY, ELECTRONIC, SPECIAL PURPOSE

The complete requirements for procuring relays described herein shall consist of this document and the latest issue of Specification MIL-R-27777.

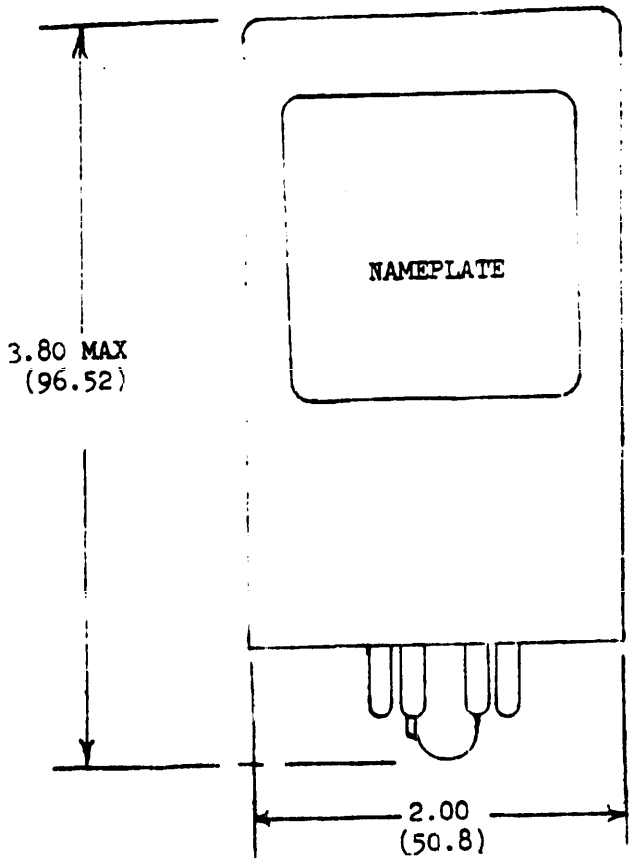


Figure 1

FSC 5945

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:52Z
Check the source to verify that this is the current version before use.

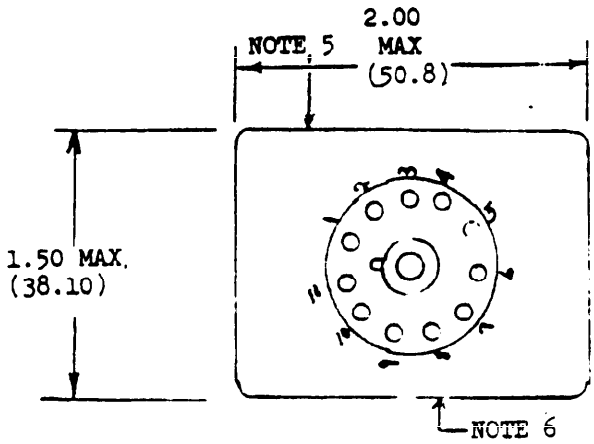


Figure 2

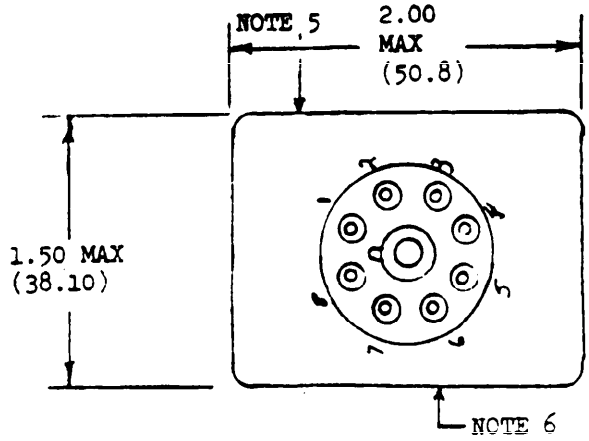


Figure 3

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4.
3. Millimeters are in parentheses.
4. Relays shall be constructed in a manner to permit wiring changes to be compatible with Repeater Group, Telegraph AN/UGA-5.
   (a) Part number M27777/4-001 shall be the input interface designated as assembly 1A1A16 of AN/UGA-5.
   (b) Part number M27777/4-002 shall be the output interface designated as assembly 1A1A17 of AN/UGA-5.
5. Marking of circuit diagram on this surface (see figures 4 and 5 as applicable).
6. Marking including part number, manufacturer's name or code and date code on this surface.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:52Z
Check the source to verify that this is the current version before use.

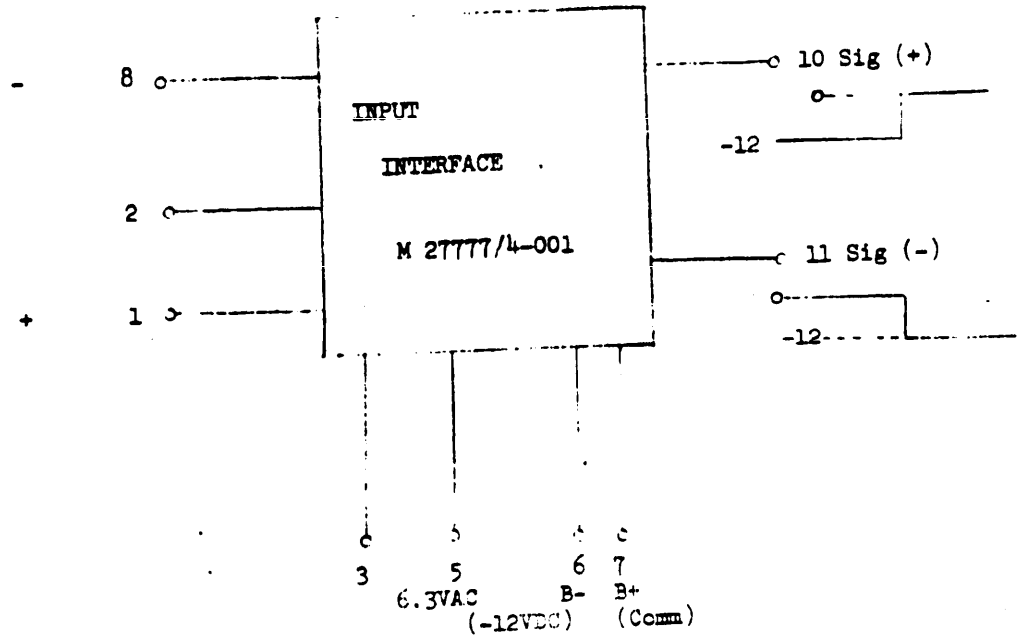


Figure 4

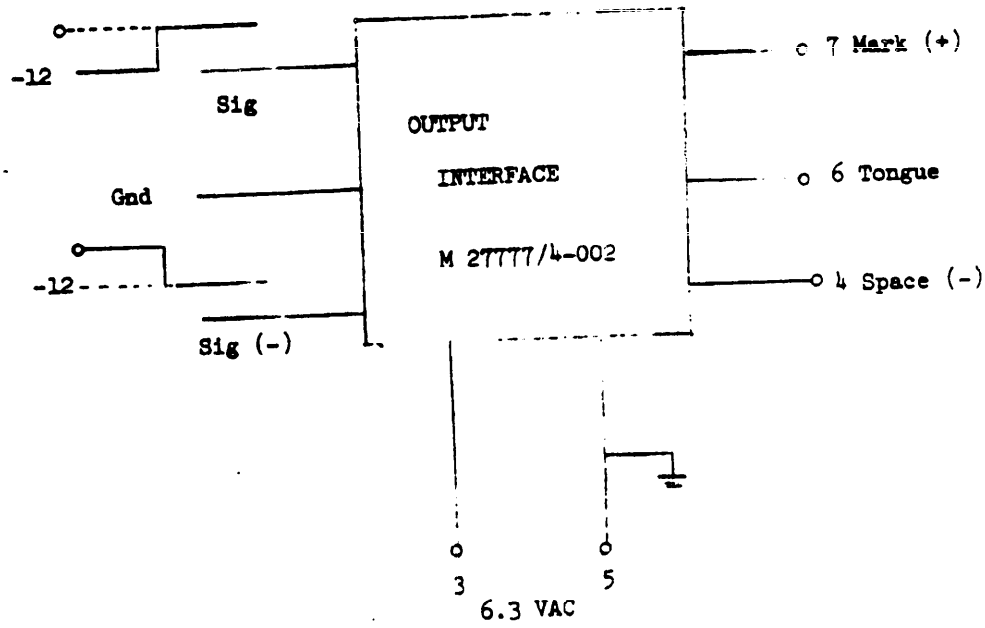


Figure 5

Circuit Diagrams

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:52Z
Check the source to verify that this is the current version before use.

CONTACT DATA:	M27777/4-001 INPUT INTERFACE	M27777/4-002 OUTPUT INTERFACE
Input	a. Polar, low level in accordance with MIL-STD-188 strap pins 2 & 8 b. Polar, current 20 or 30MA Strap pins 2 & 8 and add 300 ohms or 200 ohms across pins 1 & 8 for 20 and 30 MA loop current respectively. c. Neutral current add 100 ohms across pins 1 & 8 for 60 MA operation add 300 ohms across pins 1 & 8 for 20 MA operation	Logic, (0 to -12 vdc)
Output	Logic level (0 to -12 vdc)	Resistive load 150 MA max at ±130 vdc, Polar 150 MA max at 260 vdc max, Neutral
Keying sense	Not applicable	a. A "mark" (contact closed) condition occurs when the voltage at pin 8 is negative with respect to pin 1. b. Mark contacts are pins 7 to 6 and space contacts are pins 4 to 6.
Impedance	Not applicable	Less than 200 Ohms, less than 10 percent transition overshoot.
Isolation	Input isolated from case ground (greater than 50 megohms) and from output.	Output isolated from case ground (greater than 50 megohms) and from input.
Keying rate	300 baud maximum	300 baud maximum
ELECTRICAL DATA:		
Insulation resistance	Not applicable	Not applicable
Dielectric withstanding voltage	Not applicable	Not applicable

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:52Z
Check the source to verify that this is the current version before use.

ELECTRICAL DATA (cont'd)	M27777/4-001 INPUT INTERFACE	M27777/4-002 OUTPUT INTERFACE
Overload test	Not applicable	Withstanding transients of 150 percent of rated voltage for 100 milli-seconds
Input resistance	Not applicable	Not applicable
Contact polarity	Applicable	Applicable
Closed-contact volt-age drop and stability	Not applicable	Not applicable
Operating speed	Applicable, except for the following: (a) speed, 300 baud (b) input and output potentials as specified herein.	Applicable, except for the following: (a) speed, 300 baud (b) input and output potentials as specified herein.
Output distortion	1 percent max at 400 baud	1 percent max at 400 baud
Switch-time	Not applicable	Not applicable
Radio Frequency Interference	Class ID in accordance with MIL-STD-461	Class ID in accordance with MIL-STD-461, except that Radio Interference shall not exceed 1 uv per/meter per kHz band-width when measured at a distance of 3 ft. in any and all directions.
Power requirements	.1 amp at 6.3 vac 47 to 63 Hertz, -12 vdc at 10 MA	.1 amp at 6.3 vac 47 to 63 Hertz
ENVIRONMENTAL DATA:		
Thermal shock	Not applicable	Not applicable
High and Low Temperature	Class 4 Applicable except 0° to 50° C.	Class 4 Applicable except 0° to 50° C.
Moisture resistance	Not applicable	Not applicable
Vibration	Not applicable	Not applicable

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:52Z
Check the source to verify that this is the current version before use.

(cont'd) ENVIRONMENTAL DATA	M27777/4-001 INPUT INTERFACE	M27777/4-002 OUTPUT INTERFACE
Shock	Not applicable	Not applicable
Seal (Hermetic)	Applicable	Applicable
PHYSICAL:		
Terminal strength	5 pounds	5 pounds
Weight	10 oz. max	10 oz. max
Dimensions and configuration	See figure 1 and table I	See figure 1 and table I
Style	See table I	See table I
Marking	Applicable	Applicable
Wiring	Detail instructions shall be supplied with each relay, covering wiring changes for operation in all modes defined herein.	Detail instructions shall be supplied with each relay, covering wiring changes for operation in all modes defined herein.
Base	Standard 11 pin see figure 2	Standard octal see figure 3
Finish	Navy Gray in accordance with MIL-E-15090 Type III, Class II	Navy Gray in accordance with MIL-E-15090 Type III, Class II
LIFE TEST REQUIREMENTS	In accordance with MIL-R-27777 or as specified in the contract or order	In accordance with MIL-R-27777 or as specified in the contract or order

TABLE I

Part number	Style	Type	FSN
M27777/4-001	A	Standard 11 pin	5805-062-4847
M27777/4-002	A	Standard Octal	5805-062-4848

Preparing activity
Navy - EC
(Project 5945-N178)

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:52Z
Check the source to verify that this is the current version before use.